

## Trial Higher School Certificate Examination

2007



# Mathematics

## Extension 2

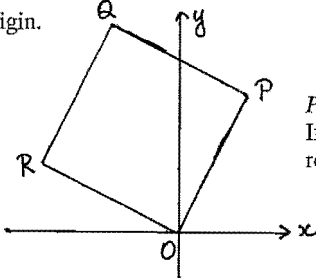
## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Student Number on every page
- All questions may be attempted.
- Begin each question in a new booklet.
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

## Question 1 – (15 marks) – Start a new booklet

Marks

- a) Evaluate  $|3 - 2i|$  1
- b) Given that  $z = x + yi$  where  $x$  and  $y$  are real numbers, express the following in the form  $a + bi$ , where  $a$  and  $b$  are real
- (i)  $(\overline{3 + i})z$  (ii)  $\frac{z}{5 - 12i}$  1, 1
- c) On an Argand diagram shade the region containing all points representing the complex number  $z$  such that:
- $$-\frac{\pi}{6} \leq \arg z < \frac{\pi}{3} \quad \text{and} \quad |z| \leq 2$$
- 3
- d) Sketch on separate diagrams the locus specified by:
- (i)  $\arg(z - (1 + i)) = \frac{\pi}{6}$  1
- (ii)  $\arg(z - 4) = \arg(z - 2i)$  2
- (iii)  $\arg\left(\frac{z - 2}{z + 2}\right) = \frac{\pi}{2}$
- e) (i) Show that multiplication of a complex number,  $z$ , by  $i$  can be represented by a rotation of the vector representing  $z$  through an angle of  $\frac{\pi}{2}$  radians about the origin. 2
- (ii)  P represents the complex number  $2 + 4i$ . If  $OPQR$  is a square, find the complex numbers represented by  $Q$  and  $R$ . 2

**Question 2 – (15 marks) – Start a new booklet**

**Marks**

- a)  $\alpha, \beta$  and  $\gamma$  are the roots of  $P(x)=0$  where  $P(x)=3x^3+7x^2+9x+1$

Find:

(i)  $\alpha^2 + \beta^2 + \gamma^2$

2

(ii)  $\alpha^3 + \beta^3 + \gamma^3$

2

- b) Find the value of  $\int_0^{\frac{2\pi}{3}} \frac{1}{5+4\cos\theta} d\theta$

4

- c) (i) Find constants  $A, B, C$  such that

$$\frac{3x^2+2x+1}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

2

- (ii) Hence, or otherwise, find  $\int \frac{3x^2+2x+1}{(x+1)(x^2+1)} dx$

2

- d) Find  $\int e^x \sin x dx$

3

**Question 3 – (15 marks) – Start a new booklet**

**Marks**

- a) The area represented by the circle  $x^2 + y^2 = a^2$  is rotated about the line  $x = a$  ( $a > 0$ ) to form a solid. Use slices to find the volume of the solid.

3

- b) When polynomial  $P(x)$  is divided by  $(x-4)$  the remainder is 3 and when  $P(x)$  is divided by  $x-3$  the remainder is 1. What is the remainder when  $P(x)$  is divided by  $x^2-7x+12$ ?

3

- c) The hyperbola,  $H$ , has equation  $xy = 9$

- (i)  $P\left(3p, \frac{3}{p}\right)$  and  $Q\left(3q, \frac{3}{q}\right)$ , where  $p > 0$  and  $q > 0$ , are 2 distinct points on  $H$ .

Show that the equation of chord  $PQ$  is  $x + pqy = 3(p+q)$ .

2

- (ii) Show that the equation of the tangent at  $P$  is  $x + p^2y = 6p$ .

2

- (iii) Find the coordinates of the point of intersection,  $T$ , of the tangents at  $P$  and  $Q$ .

2

- (iv)  $PQ$  always passes through the point  $(0, 9)$ . Find the equation of the locus of  $T$ .

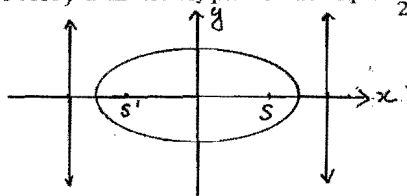
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**Question 4 – (15 marks) – Start a new booklet**

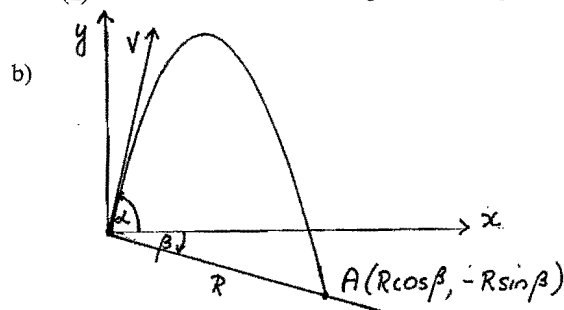
**Marks**

- a)  $P(5 \sin \theta, 3 \cos \theta)$  is an arbitrary point on the ellipse  $\frac{x^2}{25} + \frac{y^2}{9} = 1$

6



- (i) Copy the diagram into your answer booklet, giving the coordinates of the foci,  $S$  and  $S'$  and the equations of the directrices.  
(ii) Show that  $PS + PS'$  is independent of the position of  $P$ .



A ball is thrown from  $O$  with velocity  $V$  at acute angle  $\alpha$  to the horizontal. It lands at  $A(R \cos \beta, -R \sin \beta)$  on a slope inclined at acute angle  $\beta$  to the horizontal as shown above.

It is given that  $OA = R$ .

The position of the ball at time  $t$  is given by

$$x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - \frac{1}{2}gt^2$$

- (i) Show that the time taken to reach  $A$  is  $\frac{R \cos \beta}{V \cos \alpha}$   
(ii) Show that  $R = \frac{2V^2 \cos^2 \alpha}{g \cos^2 \beta} (\tan \alpha \cos \beta + \sin \beta)$   
(iii) If  $V = 10\sqrt{g}$  and  $\alpha = \beta$

1

2

- (a) find  $R$  as a function of  $\alpha$

1

- (b) with careful explanation, find the maximum value of  $R$  if  $0 < \alpha \leq \frac{\pi}{4}$

2

- c) The area bounded by the curves  $y = (x-1)^2$  and  $y = x+1$  is rotated about the  $y$ -axis to form a solid. Use cylindrical shells to find the volume of this solid.

3

**Question 5 – (15 marks) – Start a new booklet**

**Marks**

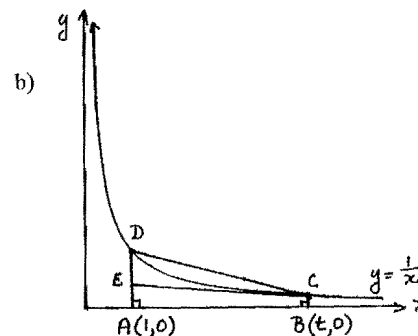
- a)  $z - ki$  ( $k$  is real) is a factor of  $P(z) = z^4 - z^3 + 9z^2 - 4z + 20$

- (i) Find possible values of  $k$

3

- (ii) Hence, or otherwise, solve  $P(z) = 0$  over the complex numbers.

3



In the above sketch of  $y = \frac{1}{x}$

- $CE$  is the tangent to  $y = \frac{1}{x}$  at  $C$
- $C$  and  $D$  lie on the curve  $y = \frac{1}{x}$

- (i) Prove that  $E$  is the point  $\left(1, \frac{2t-1}{t^2}\right)$

2

- (ii) By considering the areas of the trapezia  $ABCE$  and  $ABCD$  show that

2

$$\frac{(t-1)(3t-1)}{2t^2} < \int_{1/x}^{t/1} dx < \frac{t^2-1}{2t}$$

- (iii) Hence show that  $\frac{5}{8} < \ln 2 < \frac{3}{4}$

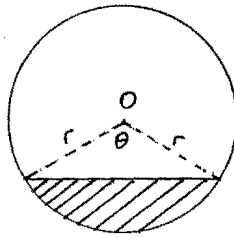
1

Part c) on next page

Question 5 (cont'd)

Marks

c)



The area of the shaded segment is one third of the area of the circle.

(i) Show that  $3\theta - 2\pi = 3 \sin \theta$  1

(ii) Carefully explain why this equation has a solution  $\theta = \alpha$  where  $\frac{\pi}{2} < \alpha < \pi$  1

(iii) With  $\alpha = \frac{3\pi}{4}$  as a first approximation, use one application of Newton's method to find a better approximation correct to 1 decimal place. 2

Question 6 – (15 marks) – Start a new booklet

Marks

- a) A body of mass one kilogram is projected vertically upwards from the ground at a speed of 20 metres per second. The particle is under the effect of both gravity and a resistance which, at any time, has a magnitude of  $\frac{1}{40}v^2$ , where  $v$  is the magnitude of the particle's velocity at that time.

In the following questions take the acceleration due to gravity to be 10 metres per second per second.

- (i) While the body is travelling upwards the equation of motion is

$$\ddot{x} = -\left(10 + \frac{1}{40}v^2\right)$$

- (a) Taking  $\dot{x} = v \frac{dv}{dx}$ , show that the greatest height reached by the particle is 20 log 2 metres 2

- (b) Taking  $\ddot{x} = \frac{dv}{dt}$ , calculate the time taken to reach this greatest height. 2

- (ii) Having reached its greatest height the particle falls to its starting point. The particle is still under the effect of both gravity and a resistance which, at any time, has a magnitude of  $\frac{1}{40}v^2$

- (a) Write down the equation of motion of the particle as it falls. 1

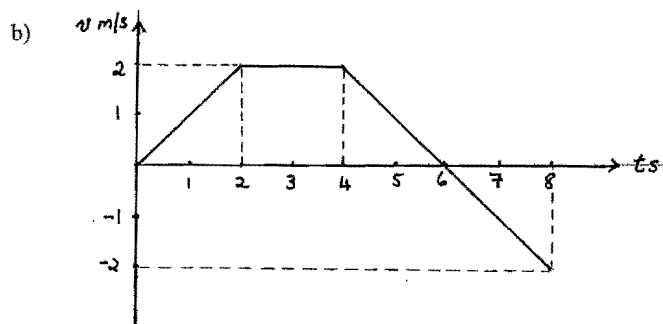
- (b) Find the speed of the particle when it returns to its starting point. 3

- (c) Express this speed as a percentage of the terminal velocity correct to 3 significant figures. 2

Part b) on next page

Question 6 (cont'd)

Marks



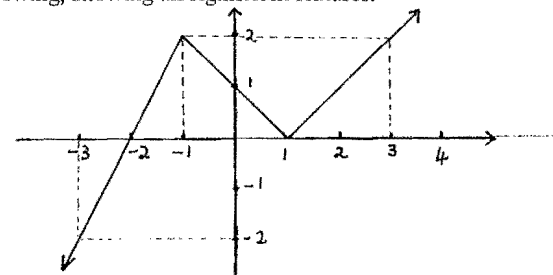
A particle of mass 2kg starts from rest at the origin and moves along the  $x$ -axis such that its velocity,  $v$  m/s at time  $t$  s where  $0 \leq t \leq 8$  is represented by the above graph.

- (i) Where is the particle at  $t = 8$  s? 2
- (ii) For the time period  $4 < t \leq 8$ , find the magnitude and direction of the resultant force on the particle and clearly describe its effect on the particle. 3

Question 7 – (15 marks) – Start a new booklet

Marks

- a) The graph of  $y = f(x)$  is shown below. On separate diagrams draw neat sketches of the following, showing all significant features.



- (i)  $y = \frac{1}{f(x)}$  2
- (ii)  $y = [f(x)]^2$  2
- (iii)  $y = \log_e |f(x)|$  2
- (iv)  $y = \tan^{-1} f(x)$  2

b)  $I_n = \int_0^1 \frac{x^n}{1+x^2} dx$  for  $n = 0, 1, 2, 3, \dots$

- (i) Show that  $I_n + I_{n+2} = \frac{1}{n+1}$  3
- (ii) Find  $I_0$  and  $I_1$  2
- (iii) Hence find  $I_6$  2

**Question 8 – (15 marks) – Start a new booklet**

Marks

a) Consider the function  $f(x) = \frac{e^x - 1}{e^x + 1}$

(i) Show that  $f(x)$  is always increasing

2

(ii) Find  $f'(0)$

1

(iii) Sketch  $y = f(x)$  showing any asymptotes

2

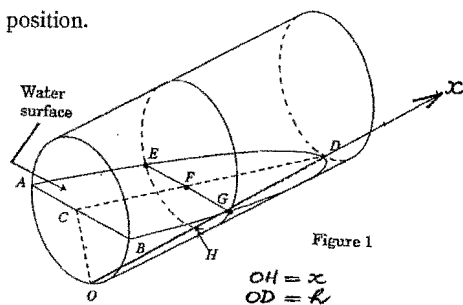
(iv) Using your graph, or otherwise, find the values of  $m$  for which

$$\frac{e^x - 1}{e^x + 1} = mx \text{ has 3 real solutions}$$

1

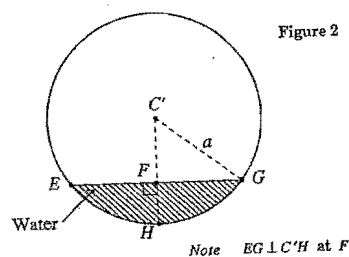
- b) A drinking glass having the form of a right circular cylinder of radius  $a$  and height  $h$ , is filled with water. The glass is slowly tilted over, spilling water out of it, until it reaches the position where the water's surface bisects the base of the glass.

Figure 1 shows this position.



In Figure 1,  $AB$  is a diameter of the circular base with centre  $C$ ,  $O$  is the lowest point on the base, and  $D$  is the point where the water's surface touches the rim of the glass.

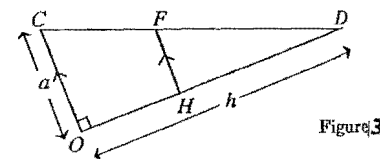
Figure 2 shows a cross-section of the tilted glass parallel to its base. The centre of this circular section is  $C'$  and  $EFG$  shows the water level. The section cuts the lines  $CD$  and  $OD$  of Figure 1 in  $F$  and  $H$  respectively.



**Question 8 (cont'd)**

Marks

Figure 3 shows the section  $COD$  of the tilted glass.



Note  $FH \parallel CO$ ,  $CO = a$ , and  $OD = h$

(i) Use Figure 3 to show that  $FH = \frac{a}{h}(h - x)$ , where  $OH = x$

1

(ii) Use Figure 2 to show that  $C'F = \frac{ax}{h}$  and  $\angle HC'G = \cos^{-1}\left(\frac{x}{h}\right)$

2

(iii) Use (ii) to show that the area of the shaded segment  $EGH$  is

$$a^2 \left[ \cos^{-1}\left(\frac{x}{h}\right) - \left(\frac{x}{h}\right) \sqrt{1 - \left(\frac{x}{h}\right)^2} \right]$$

2

(iv) Given that  $\int \cos^{-1} \theta d\theta = \theta \cos^{-1} \theta - \sqrt{1 - \theta^2}$ , find the volume of water in the tilted glass of Figure 1.

4

**End of Paper**

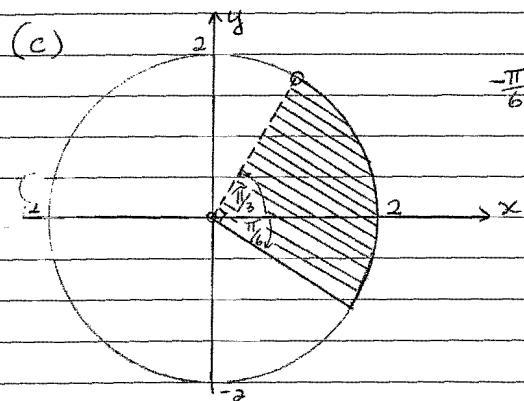
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Question 1

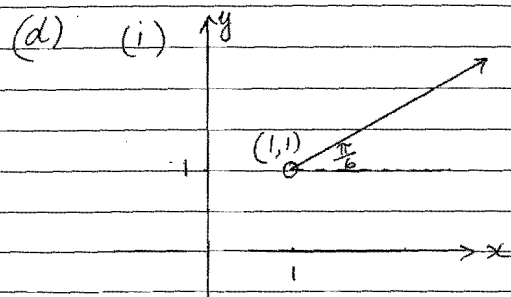
$$(a) |3-2i| = \sqrt{3^2 + (-2)^2} \\ = \sqrt{13}$$

$$(b) (i) (3+i)z = (3-i)(x+yi) \\ = 3x+y + (3y-x)i$$

$$(ii) \frac{z}{5-12i} = \frac{(x+yi)(5+12i)}{(5-12i)(5+12i)} \\ = \frac{5x-12y + i(12x+5y)}{5^2+12^2} \\ = \frac{5x-12y}{169} + \frac{12x+5y}{169}i$$

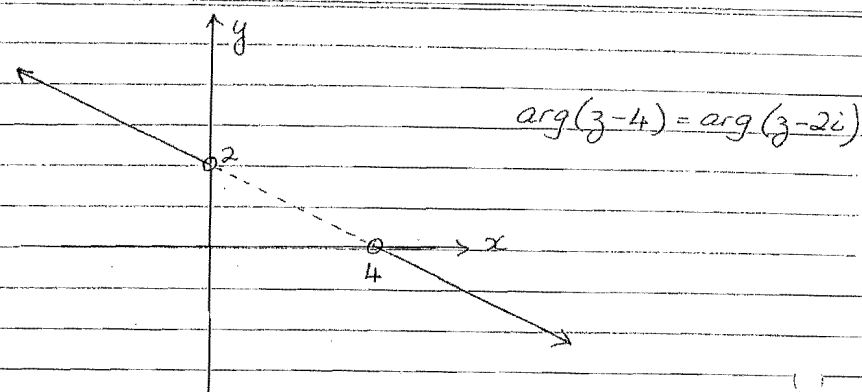


$$-\frac{\pi}{6} \leq \arg z < \frac{\pi}{3} \text{ and } |z| \leq 2$$

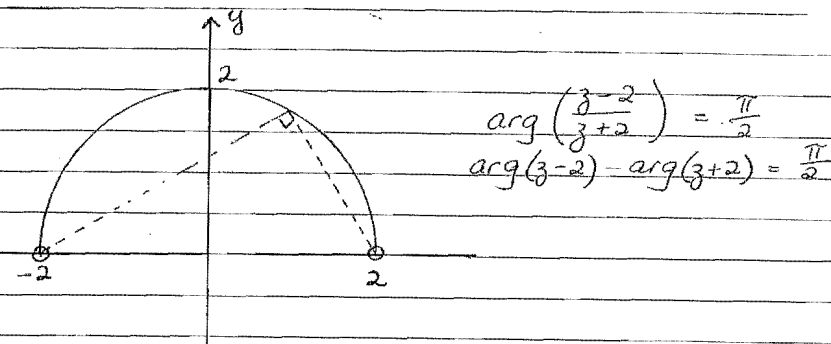


$$\arg(z - (1+i)) = \frac{\pi}{6}$$

(d) (ii)



(iii)



e) (i) Let  $OP$ ,  $OQ$  be the position vectors representing  $z$  and  $iz$  on the Argand diagram

$$OP = |z| \quad OQ = |iz| = |i||z| = 1 \cdot |z| = OP$$

$$\arg(iz) = \arg i + \arg z = \frac{\pi}{2} + \arg z$$

ie  $OQ$  makes an angle of  $\frac{\pi}{2}$  with  $OP$

ie The transformation  $z \rightarrow iz$  is a rotation about  $O$  through  $\frac{\pi}{2}$  radians (in the anticlockwise direction)

(ii)  $R$  represents  $i(2+4i) = -4+2i$

$$Q \text{ represents } (2+4i) + (-4+2i) = -2+6i$$

## Question 2

(a)  $P(x) = 3x^3 + 7x^2 + 9x + 1$

(i)  $\alpha + \beta + \gamma = -\frac{7}{3}$   
 $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{9}{3} = 3$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= \left(-\frac{7}{3}\right)^2 - 2 \times 3 \\ &= \frac{49}{9} - 6 \\ &= -\frac{5}{9}\end{aligned}$$

(ii)  $\begin{cases} 3\alpha^3 + 7\alpha^2 + 9\alpha + 1 = 0 \\ 3\beta^3 + 7\beta^2 + 9\beta + 1 = 0 \\ 3\gamma^3 + 7\gamma^2 + 9\gamma + 1 = 0 \end{cases}$  since  $\alpha, \beta, \gamma$  are roots of  $P(x) = 0$

$$\begin{aligned}\therefore 3(\alpha^3 + \beta^3 + \gamma^3) + 7(\alpha^2 + \beta^2 + \gamma^2) + 9(\alpha + \beta + \gamma) + 3 &= 0 \\ 3(\alpha^3 + \beta^3 + \gamma^3) + 7 \times -\frac{5}{9} + 9 \times -\frac{7}{3} + 3 &= 0\end{aligned}$$

$$3(\alpha^3 + \beta^3 + \gamma^3) = \frac{35}{9} + 21 - 3$$

$$\alpha^3 + \beta^3 + \gamma^3 = \frac{78}{27} \quad \left( = \frac{197}{27} \right)$$

(b)  $\int_0^{\frac{\pi}{2}} \frac{1}{5+4\cos\theta} d\theta$

$$= \int_0^{\sqrt{3}} \frac{1}{5+4\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int_0^{\sqrt{3}} \frac{2}{5+5t^2+4-4t^2} dt$$

$$= \int_0^{\sqrt{3}} \frac{2}{9+t^2} dt$$

$$t = \tan \frac{\theta}{2}$$

$$\theta = 2 \tan^{-1} t$$

$$d\theta = \frac{2}{1+t^2} dt$$

When  $\theta = 0$   $t = 0$

$$\theta = \frac{\pi}{3} \quad t = \sqrt{3}$$

2(b) (cont)  $\int_0^{\sqrt{3}} \frac{2}{9+t^2} dt = \frac{2}{3} \left[ \tan^{-1} \frac{t}{3} \right]_0^{\sqrt{3}}$   
 $= \frac{2}{3} \left( \tan^{-1} \frac{\sqrt{3}}{3} - \tan^{-1} 0 \right)$   
 $= \frac{2}{3} \left( \frac{\pi}{6} - 0 \right)$   
 $= \frac{\pi}{9}$

(c)  $\frac{3x^2+2x+1}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$

$$3x^2+2x+1 \equiv A(x^2+1) + (Bx+C)(x+1)$$

$x = -1 \quad 3 - 2 + 1 = 2A$

$$A = 1$$

Coeff  $x^2$ :  $3 = A + B$

$$B = 2$$

$x = 0 \quad 1 = A + C$

$$C = 0$$

(ii)  $\int \frac{3x^2+2x+1}{(x+1)(x^2+1)} dx = \int \frac{1}{x+1} + \frac{2x}{x^2+1} dx$

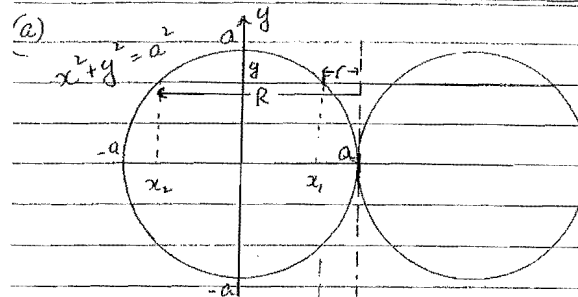
$$= \ln|x+1| + \ln(x^2+1) + C$$

$$\begin{aligned}
 (d) \int e^x \sin x dx &= \int \frac{d(e^x)}{dx} \sin x dx \\
 &= e^x \sin x - \int e^x \cos x dx \\
 &= e^x \sin x - \int \frac{d(e^x)}{dx} \cos x dx \\
 &= e^x \sin x - \left\{ e^x \cos x - \int e^x (-\sin x) dx \right\} \\
 &= e^x \sin x - e^x \cos x - \int e^x \sin x dx
 \end{aligned}$$

$$2 \int e^x \sin x dx = e^x (\sin x - \cos x) + C$$

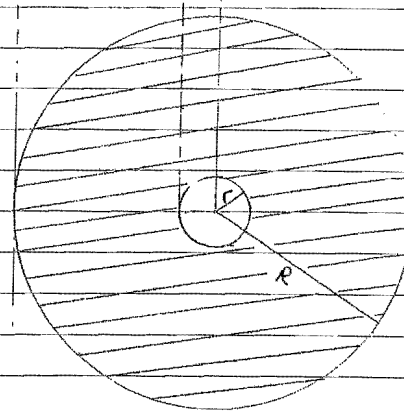
$$\int e^x \sin x dx = \frac{1}{2} e^x (\sin x - \cos x) + C$$

### Question 3



$$\begin{aligned}
 x_1 &= \sqrt{a^2 - y^2} \\
 x_2 &= -\sqrt{a^2 - y^2}
 \end{aligned}$$

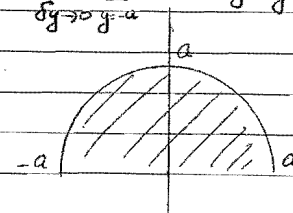
$$\begin{aligned}
 R &= a + \sqrt{a^2 - y^2} \\
 r &= a - \sqrt{a^2 - y^2}
 \end{aligned}$$



$$\begin{aligned}
 A(y) &= \pi R^2 - \pi r^2 \\
 &= \pi (R-r)(R+r) \\
 &= \pi (2\sqrt{a^2 - y^2})(2a) \\
 &= 4a\pi \sqrt{a^2 - y^2}
 \end{aligned}$$

$$\delta V \doteq A(y) \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=a}^a 4a\pi \sqrt{a^2 - y^2} \delta y$$



$$\begin{aligned}
 V &= \int_{-a}^a A(y) dy \\
 &= 4a\pi \int_{-a}^a \sqrt{a^2 - y^2} dy \\
 &= 4a\pi \times \frac{1}{2} \cdot \pi a^2 \\
 &= 2a^3 \pi^2
 \end{aligned}$$

$$(4) \quad P(4) = 3 \quad P(3) = 1$$

$$\text{Let } P(x) = (x^2 - 7x + 12)Q(x) + ax + b \\ = (x-4)(x-3)Q(x) + ax + b$$

$$P(4) = 4a + b = 3 \quad \textcircled{1}$$

$$P(3) = 3a + b = 1 \quad \textcircled{2}$$

$$\textcircled{1} - \textcircled{2} \quad a = 2$$

$$\therefore 6 + b = 1$$

$$b = -5$$

$\therefore$  Remainder is  $2x - 5$

$$(c) \quad H: xy = 9 \quad P(3p, \frac{3}{p}) \quad Q(3q, \frac{3}{q})$$

$$(i) \quad \text{Grad } PQ = \frac{\frac{3}{p} - \frac{3}{q}}{\frac{3p}{p^2} - \frac{3q}{q^2}} \\ = \frac{3(q-p)}{pq} \times \frac{1}{3(p-q)} \\ = -\frac{1}{pq}$$

$$\therefore \text{Eqn of } PQ \text{ is } y - \frac{3}{p} = -\frac{1}{pq}(x - 3p)$$

$$pqy - 3q = -x + 3p$$

$$x + pqy = 3p + 3q \\ = 3(p+q)$$

$$(ii) \quad y = \frac{9}{x} \\ \frac{dy}{dx} = -9x^{-2}$$

$$\text{At } P \quad \frac{dy}{dx} = \frac{9}{9p^2} = \frac{-1}{p^2}$$

$\therefore$  Eqn of tangent at P is

$$y - \frac{3}{p} = -\frac{1}{p^2}(x - 3p)$$

$$p^2y - 3p = -x + 3p$$

$$x + p^2y = 6p$$

(iii) Tangents at P and Q:

$$x + p^2y = 6p \quad \textcircled{1}$$

$$x + q^2y = 6q \quad \textcircled{2}$$

Meet at T:

$$\textcircled{1} - \textcircled{2} \quad (p^2 - q^2)y = 6(p - q)$$

$$(p - q)(p + q)y = 6(p - q)$$

$$y = \frac{6}{p + q} \quad (p \neq q)$$

$$x = 6p - p^2 \times \frac{6}{p + q}$$

$$= \frac{6p^2 + 6pq - 6p^2}{p + q}$$

$$= \frac{6pq}{p + q}$$

$\therefore$  T has coordinates  $(\frac{6pq}{p+q}, \frac{6}{p+q})$

(iv) Since PQ passes through (0, 9)

$$0 + 9pq = 3(p+q)$$

$$3pq = p+q$$

For T:  $x = \frac{6pq}{p+q}$   $y = \frac{6}{p+q}$

$$= \frac{2 \cdot 3pq}{p+q}$$

$$= \frac{2(p+q)}{p+q}$$

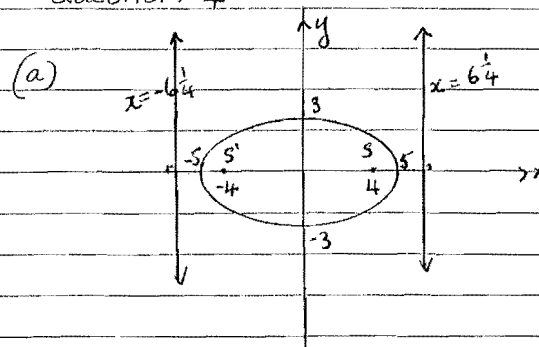
$$= 2$$

Since  $p > 0$  and  $q > 0$   $p+q > 0$  and so  $y > 0$

Tangents intersect below the curve: When  $x=2$   $y = \frac{9}{2}$

$\therefore$  Locus of T is  $x=2$   $0 < y < \frac{9}{2}$

Question 4



$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$a^2 = 25$$

$$b^2 = 9$$

$$= a^2(1-e^2)$$

$$9 = 25(1-e^2)$$

$$e^2 = 1 - \frac{9}{25}$$

$$= \frac{16}{25}$$

$$e = \frac{4}{5} \quad (e > 0)$$

$$ae = 5 \times \frac{4}{5} = 4$$

$$\frac{a}{e} = \frac{5}{\frac{4}{5}} = \frac{25}{4}$$

$$S(4,0) \quad S'(-4,0)$$

$$\text{Directrices } x = \pm \frac{25}{4}$$

(ii) Let M, M' be feet of perpendiculars from P to the directrices

$$PS = e PM = \frac{4}{5} PM$$

$$PS' = e PM' = \frac{4}{5} PM'$$

$$PS + PS' = \frac{4}{5} (PM + PM')$$

$$= \frac{4}{5} \times MM'$$

$$= \frac{4}{5} \times 2 \times \frac{25}{4} \quad (1)$$

$$= 10$$

ie  $PS + PS'$  is independent of P.

$$(k) \quad x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - \frac{1}{2} g t^2$$

$$(i) \quad \text{At A } x = R \cos \beta$$

$$\therefore R \cos \beta = Vt \cos \alpha$$

$$\therefore t = \frac{R \cos \beta}{V \cos \alpha}$$

$$(ii) \quad -R \sin \beta = V \cdot \frac{R \cos \beta}{V \cos \alpha} \cdot \sin \alpha - \frac{1}{2} g \cdot \frac{R^2 \cos^2 \beta}{V^2 \cos^2 \alpha}$$

$$= R \tan \alpha \cos \beta - \frac{g R^2 \cos^2 \beta}{2 V^2 \cos^2 \alpha} \quad (\div R)$$

$$\frac{R g \cos^2 \beta}{2 V^2 \cos^2 \alpha} = \tan \alpha \cos \beta + \sin \beta$$

$$R = \frac{2 V^2 \cos^2 \alpha}{g \cos^2 \beta} (\tan \alpha \cos \beta + \sin \beta)$$

$$(iii) \quad \text{If } V = 10\sqrt{g} \quad \alpha = \beta$$

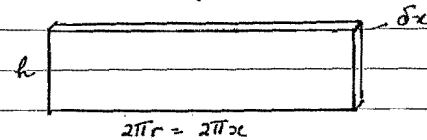
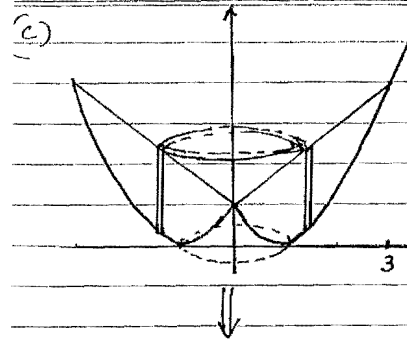
$$(a) \quad R = \frac{2 \times 100g \cos^2 \alpha}{g \cos^2 \alpha} (\tan \alpha \cos \alpha + \sin \alpha)$$

$$= 200 (\sin \alpha + \sin \alpha)$$

$$= 400 \sin \alpha$$

(4) Since  $\sin \alpha$  is an increasing function for  $0 < \alpha \leq \frac{\pi}{4}$ , maximum value of  $R$  for this domain occurs when  $\alpha = \frac{\pi}{4}$

$$\therefore \text{Max } R = 400 \sin \frac{\pi}{4} = 400 \times \frac{1}{\sqrt{2}} = 200\sqrt{2}$$



$$y = x+1 \text{ meets } y = (x-1)^2$$

$$x^2 - 2x + 1 = x + 1$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$x = 0, 3$$

$$r = x$$

$$h = (x+1) - (x-1)^2$$

$$= 3x - x^2$$

$$\delta V \doteq 2\pi r h \delta x$$

$$= 2\pi x (3x - x^2) \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 2\pi x (3x - x^2) \delta x$$

$$= \int_0^3 2\pi x (3x - x^2) dx$$

$$= 2\pi \int_0^3 (3x^2 - x^3) dx$$

$$= 2\pi \left[ x^3 - \frac{x^4}{4} \right]_0^3$$

$$= 2\pi \left\{ \left( 27 - \frac{81}{4} \right) - 0 \right\}$$

$$= \frac{27\pi}{2}$$

### Question 5

$$(a) P(z) = z^4 - z^3 + 9z^2 - 4z + 20$$

$$(i) P(ki) = 0$$

$$(ki)^4 - (ki)^3 + 9(ki)^2 - 4(ki) + 20 = 0$$

$$k^4 - k^3(-i) + 9(-k^2) - 4ki + 20 = 0$$

Equating real parts gives

$$k^4 - 9k^2 + 20 = 0$$

$$(k^2 - 4)(k^2 - 5) = 0$$

$$(k-2)(k+2)(k-\sqrt{5})(k+\sqrt{5}) = 0$$

$$k = 2, -2, \sqrt{5}, -\sqrt{5}$$

Equating imaginary parts gives

$$k^3 - 4k = 0$$

$$k(k^2 - 4) = 0$$

$$k = 0, 2, -2$$

$$\therefore \{2, -2, \sqrt{5}, -\sqrt{5}\} \cap \{0, 2, -2\} = \{2, -2\}$$

$$\therefore k = 2, -2$$

$$(ii) \therefore (z-2i)(z+2i) \mid P(z)$$

$$\therefore z^4 - z^3 + 9z^2 - 4z + 20 = (z^2 + 4)(z^2 - z + 5)$$

$$\therefore \text{If } P(z) = 0$$

$$z = \pm 2i, \frac{1 \pm \sqrt{1-4 \times 1 \times 5}}{2}$$

$$= \pm 2i, \frac{1 \pm i\sqrt{19}}{2}$$

$$(h) (i) y = \frac{1}{x}$$

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

$$\text{At } C(t, \frac{1}{t}) \quad \frac{dy}{dx} = -\frac{1}{t^2}$$

Eq<sup>n</sup> of tangent at C is

$$y - \frac{1}{t} = -\frac{1}{t^2}(x - t)$$

$$t^2 y - t = -x + t$$

$$t^2 y = -x + 2t$$

$$\text{When } x=1 \quad t^2 y = -1 + 2t$$

$$y = \frac{2t-1}{t^2}$$

$$\therefore E \text{ has coords } (1, \frac{2t-1}{t^2})$$

$$(ii) \text{Area } ABCE = \frac{(t-1)}{2} \times \left( \frac{2t-1}{t^2} + \frac{1}{t} \right) = \left[ \frac{AB(AE+BC)}{2} \right]$$

$$= \frac{(t-1)}{2} \times \frac{(2t-1+t)}{t^2}$$

$$= \frac{(t-1)(3t-1)}{2t^2}$$

$$\text{Area } ABCD = \frac{(t-1)}{2} \left( 1 + \frac{1}{t} \right) = \left[ \frac{AB(AD+BC)}{2} \right]$$

$$= \frac{(t-1)(t+1)}{2t}$$

$$= \frac{t^2-1}{2t}$$

$$\text{Area under curve from D to C} = \int_1^t \frac{1}{x} dx$$

Area of ABCE < Area under curve < area of ABCD

$$\frac{(t-1)(3t-1)}{2t^2} < \int_1^t \frac{1}{x} dx < \frac{t^2-1}{2t}$$

(iii) Let  $t = 2$

$$\frac{(2-1)(3 \times 2-1)}{2 \times 2^2} < \int_1^2 \frac{1}{x} dx < \frac{2^2-1}{2 \times 2}$$

$$\frac{5}{8} < [\ln x]^2 < \frac{3}{4}$$

$$\frac{5}{8} < \ln 2 - \ln 1 < \frac{3}{4}$$

$$\frac{5}{8} < \ln 2 < \frac{3}{4}$$

(c) (i) Shaded area =  $\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$

$$\therefore \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta = \frac{1}{3}\pi r^2 \quad \left(\times \frac{6}{r^2}\right)$$

$$3\theta - 3\sin\theta = 2\pi$$

$$3\theta - 2\pi = 3\sin\theta$$

(ii) Let  $f(\theta) = 3\theta - 2\pi - 3\sin\theta$

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \frac{3\pi}{2} - 2\pi - 3\sin\frac{\pi}{2} \\ &= -\frac{\pi}{2} - 3 \\ &< 0 \end{aligned}$$

$$\begin{aligned} f(\pi) &= 3\pi - 2\pi - 3\sin\pi \\ &= \pi - 0 \\ &= \pi \\ &> 0 \end{aligned}$$

$\therefore$  Since  $f\left(\frac{\pi}{2}\right) < 0$ ,  $f(\pi) > 0$  and  $f(\theta)$  is continuous  $f(\theta) = 0$  has a solution  $\theta = \alpha$  where  $\frac{\pi}{2} < \alpha < \pi$

(iii)  $f(\theta) = 3\theta - 2\pi - 3\sin\theta$   
 $f'(\theta) = 3 - 3\cos\theta$

$$\begin{aligned} f\left(\frac{3\pi}{4}\right) &= \frac{9\pi}{4} - 2\pi - 3\sin\frac{3\pi}{4} \\ &= \frac{\pi}{4} - \frac{3}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} f'\left(\frac{3\pi}{4}\right) &= 3 - 3\cos\frac{3\pi}{4} \\ &= 3 + \frac{3}{\sqrt{2}} \end{aligned}$$

Let  $x_1$  be the second approximation

$$x_1 = x - \frac{f(x)}{f'(x)}$$

$$= \frac{3\pi}{4} - \frac{\frac{\pi}{4} - \frac{3}{\sqrt{2}}}{3 + \frac{3}{\sqrt{2}}}$$

$$= 2.61704\dots$$

$$= 2.6 \text{ (1 decimal place)}$$

Question 6

$$(a) \quad \ddot{x} = -\left(10 + \frac{v^2}{40}\right)$$

$$v \frac{dv}{dx} = -\left(10 + \frac{v^2}{40}\right)$$

$$\frac{dv}{dx} = -\frac{(400 + v^2)}{40v}$$

$$\frac{dx}{dv} = -\frac{40v}{400 + v^2}$$

Let  $H$  be the greatest height

$$H = \int_{20}^0 -\frac{40v}{400 + v^2} dv$$

$$= 20 \int_0^{20} \frac{2v}{400 + v^2} dv$$

$$= 20 \left[ \log_e (400 + v^2) \right]_0^{20}$$

$$= 20 \{ \log_e 800 - \log_e 400 \}$$

$$= 20 \log_e 2$$

$\therefore$  Greatest height is  $20 \log_e 2$

$$(b) \quad \frac{dv}{dt} = -\left(10 + \frac{v^2}{40}\right)$$

$$\frac{dt}{dv} = -\frac{40}{400 + v^2}$$

Let  $T$  be the time taken to reach  $H$

$$T = \int_{20}^0 -\frac{40}{400 + v^2} dv$$

$$\begin{aligned}
 T &= \int_0^{20} \frac{40}{400+v^2} dv \\
 &= \frac{40}{20} \left[ \tan^{-1} \frac{v}{20} \right]_0^{20} \\
 &= 2 \{ \tan^{-1} 1 - \tan^{-1} 0 \} \\
 &= 2 \left\{ \frac{\pi}{4} - 0 \right\} \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$\therefore$  Time taken is  $\frac{\pi}{2}$  seconds

$$\begin{aligned}
 \text{(ii) (a)} \quad \ddot{x} &= g - \frac{1}{40} v^2 \\
 &= 10 - \frac{v^2}{40} \\
 &= \frac{400 - v^2}{40}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad v \frac{dv}{dx} &= \frac{400 - v^2}{40} \\
 \frac{dv}{dx} &= \frac{400 - v^2}{40v} \\
 \frac{dx}{dv} &= \frac{40v}{400 - v^2}
 \end{aligned}$$

$$\begin{aligned}
 H &= \int_0^V \frac{40v}{400 - v^2} dv \quad \text{where } V \text{ is speed wh} \\
 &\quad \text{returns to starting} \\
 20 \log_e 2 &= -20 \int_0^V \frac{-2v}{400 - v^2} dv \quad \text{point} \\
 &= -20 \left[ \log_e (400 - v^2) \right]_0^V \\
 &= -20 \{ \log_e (400 - v^2) - \log_e 400 \}
 \end{aligned}$$

$$20 \log_e 2 = 20 \log_e \frac{400}{400 - v^2}$$

$$2 = \frac{400}{400 - v^2}$$

$$800 - 2v^2 = 400$$

$$2v^2 = 400$$

$$v^2 = 200$$

$$v = 10\sqrt{2}$$

$$\text{(c)} \quad \ddot{x} = 10 - \frac{v^2}{40}$$

Terminal velocity,  $V_T$ , occurs as  $\ddot{x} \rightarrow 0$

$$\text{ie } 10 - \frac{V_T^2}{40} = 0$$

$$V_T^2 = 400$$

$$V_T = 20$$

$$\frac{V}{V_T} = \frac{10\sqrt{2}}{20} \times 100$$

$$= \frac{\sqrt{2}}{2}$$

$$V = \frac{\sqrt{2}}{2} V_T$$

$$= 70.7\% \text{ of } V_T$$

(4) (i)  $x = 6$

(ii)  $\ddot{x} = -1 \text{ (ms}^{-2}\text{)}$

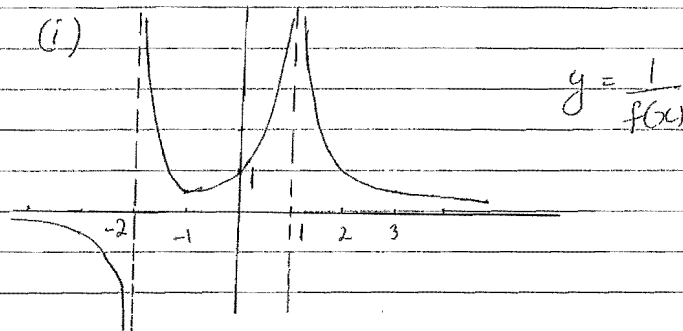
$m\ddot{x} = 2x - 1 \text{ N}$

$\therefore$  Resultant force is 2 N in negative direction

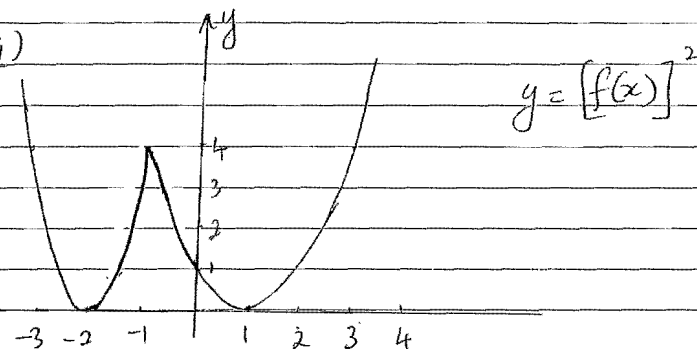
Slows particle down from 2 m/s ( $t=4$ ) until it stops, <sup>at  $t=6$</sup>  turns around and moves in negative direction at increasing speed.

Question 7.

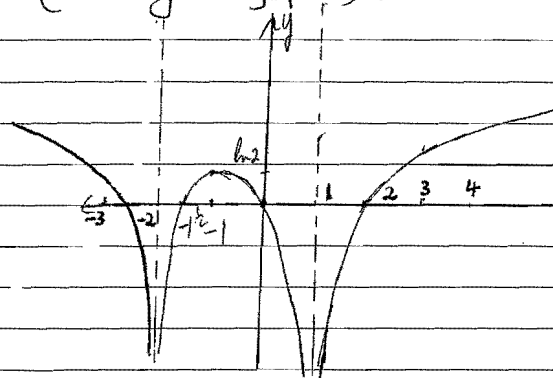
(a) (i)



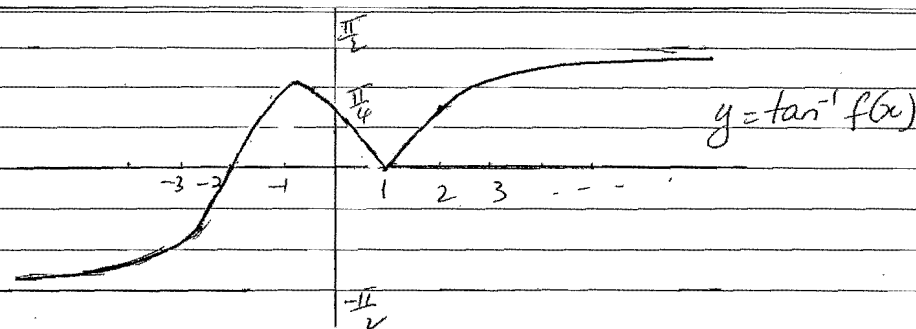
(ii)



(iii)  $y = \log_e |f(x)|$



(iv)



(4)

$$I_n = \int_0^1 \frac{x^n}{1+x^2} dx$$

$$I_n + I_{n+2} = \int_0^1 \frac{x^n}{1+x^2} + \frac{x^{n+2}}{1+x^2} dx$$

$$= \int_0^1 \frac{x^n(1+x^2)}{1+x^2} dx$$

$$= \int_0^1 x^n dx$$

$$= \left[ \frac{x^{n+1}}{n+1} \right]_0^1$$

$$= \left( \frac{1}{n+1} - \frac{0}{n+1} \right)$$

$$= \frac{1}{n+1}$$

$$(ii) \quad I_0 = \int_0^1 \frac{1}{1+x^2} dx$$

$$= \left[ \tan^{-1} x \right]_0^1$$

$$= \tan^{-1} 1 - \tan^{-1} 0$$

$$= \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

$$I_1 = \int_0^1 \frac{x}{1+x^2} dx$$

$$= \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$$

$$= \frac{1}{2} \left[ \ln(1+x^2) \right]_0^1$$

$$= \frac{1}{2} (\ln 2 - \ln 1)$$

$$= \frac{1}{2} \ln 2$$

$$(ii) \quad I_0 + I_2 = \frac{1}{1}$$

$$I_2 = 1 - \frac{\pi}{4}$$

$$I_1 + I_4 = \frac{1}{3}$$

$$I_4 = \frac{1}{3} - \left( 1 - \frac{\pi}{4} \right)$$

$$= \frac{\pi}{4} - \frac{2}{3}$$

$$I_4 + I_6 = \frac{1}{5}$$

$$I_6 = \frac{1}{5} - \left( \frac{\pi}{4} - \frac{2}{3} \right)$$

$$= \frac{13}{15} - \frac{\pi}{4}$$

### Question 8

$$(a) f(x) = \frac{e^x - 1}{e^x + 1}$$

$$(i) f'(x) = \frac{e^x(e^x + 1) - e^x(e^x - 1)}{(e^x + 1)^2}$$

$$= \frac{2e^x}{(e^x + 1)^2}$$

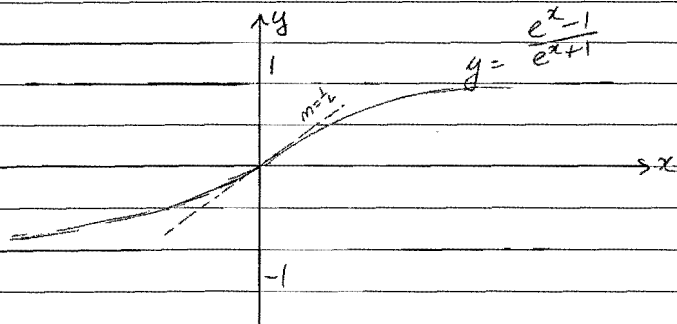
$$> 0 \quad \forall x \text{ since } e^x > 0$$

$\therefore f(x)$  is <sup>always</sup> increasing

$$(ii) f'(0) = \frac{2e^0}{(e^0 + 1)^2} = \frac{2}{(1+1)^2} = \frac{1}{2}$$

$$(iii) \text{ As } x \rightarrow \infty, f(x) \rightarrow 1^-$$

$$x \rightarrow -\infty, f(x) \rightarrow -1^+$$



$$(iv) 3 \text{ real solutions whenever } 0 < m < \frac{1}{2}$$

$$(i) \frac{FH}{CO} = \frac{DH}{DO} \quad (\text{corresp sides in similar } \Delta s \text{ FHD} \sim \text{COD})$$

$$\frac{FH}{a} = \frac{h-x}{h}$$

$$FH = \frac{a(h-x)}{h}$$

$$(ii) C'F = a - FH$$

$$= a - \frac{a(h-x)}{h}$$

$$= \frac{ah - ah + ax}{h}$$

$$= \frac{ax}{h}$$

$$\cos(\angle HC'G) = \frac{C'F}{C'G}$$

$$= \frac{\frac{ax}{h}}{\frac{1}{a}}$$

$$= \frac{x}{h}$$

$$\angle HC'G = \cos^{-1}\left(\frac{x}{h}\right)$$

(iii)

$$\text{Area of } EGH = \frac{1}{2} a^2 \theta - \frac{1}{2} a^2 \sin \theta \quad \text{where } \theta = 2 \times \angle HC'G$$

$$= \frac{1}{2} a^2 \cdot 2 \cos^{-1}\left(\frac{x}{h}\right) - \frac{1}{2} a^2 \cdot 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$= a^2 \cos^{-1}\left(\frac{x}{h}\right) - a^2 \sqrt{1 - \left(\frac{x}{h}\right)^2} \cdot \frac{x}{h}$$

$$= a^2 \left[ \cos^{-1}\left(\frac{x}{h}\right) - \frac{x}{h} \sqrt{1 - \left(\frac{x}{h}\right)^2} \right]$$

$$(IV) \quad \text{Volume} = \int_0^h a^2 \left[ \cos^{-1}\left(\frac{x}{a}\right) - \frac{x}{a} \sqrt{1 - \left(\frac{x}{a}\right)^2} \right] dx$$

$$\text{Let } \theta = \frac{x}{a} \quad dx = a d\theta$$

$$\text{When } x=0 \quad \theta=0$$

$$x=a \quad \theta=1$$

$$\begin{aligned} V &= \int_0^1 a^2 (\cos^{-1} \theta - \theta \sqrt{1-\theta^2}) a d\theta \\ &= a^3 \left[ \theta \cos^{-1} \theta - \sqrt{1-\theta^2} + \frac{1}{2} (1-\theta^2)^{3/2} \right]_0^1 \\ &= a^3 \left\{ (1 \cos^{-1} 1 - 0 + 0) - (0 - 1 + \frac{1}{2} \times 1) \right\} \\ &= a^3 \left\{ 0 - 0 + 0 + \left(\frac{1}{2}\right) \right\} \\ &= \frac{2a^3}{3} \end{aligned}$$